

# Password Strength Evaluation Report

## Objective

Understand what makes a password strong by evaluating multiple passwords of varying complexity against common password strength criteria (similar to tools like PasswordMeter) and summarizing best practices.

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## Methodology

- Created passwords with increasing complexity (length, character sets, randomness).
  - Evaluated each against common strength rules:
    - Length (8–16+ characters)
    - Uppercase, lowercase, numbers, symbols
    - Penalties for repetition, sequences, dictionary words
  - Recorded estimated score (0–100) and qualitative feedback.
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## Password Tests & Results

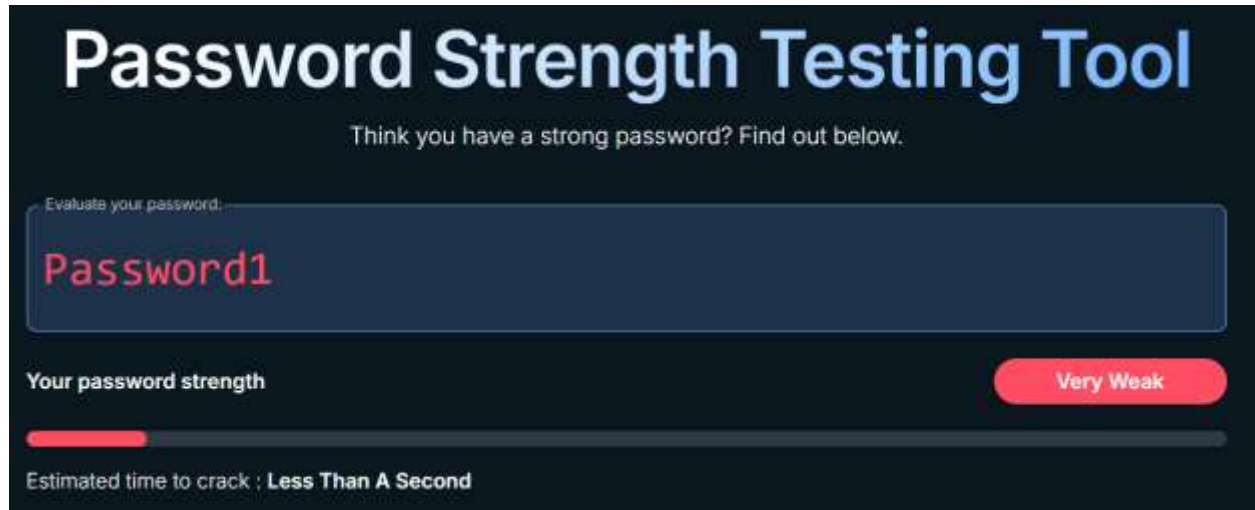
### ★ password

- **Strength Rating:** Very Weak
- **Estimated Crack Time:** Less than a second
- **Explanation:** This password is a common dictionary word with no uppercase letters, numbers, symbols, or sufficient length. Attackers include it in the very first attempts of dictionary and brute-force attacks, making it extremely insecure.



### ❖ Password1

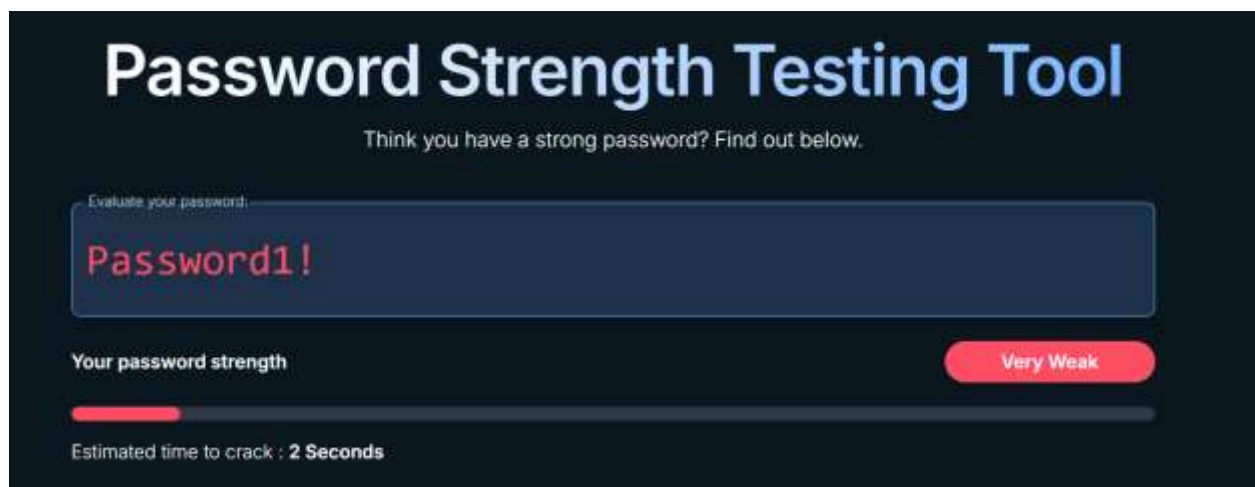
- **Strength Rating:** Very Weak
- **Estimated Crack Time:** Less than a second
- **Explanation:** Although it includes an uppercase letter and a number, the structure follows a highly predictable pattern (capitalized first letter + number at the end). Password cracking tools are specifically designed to guess such variations quickly.



The screenshot shows the 'Password Strength Testing Tool' interface. At the top, the title 'Password Strength Testing Tool' is displayed in large blue letters. Below it, a subtitle reads 'Think you have a strong password? Find out below.' A text input field contains the password 'Password1' in red text. Below the input field, the text 'Your password strength' is followed by a red progress bar that is only about 10% full. To the right of the progress bar, a red button labeled 'Very Weak' is visible. At the bottom, the text 'Estimated time to crack : Less Than A Second' is displayed.

### ❖ Password1!

- **Strength Rating:** Very Weak
- **Estimated Crack Time:** 2 seconds
- **Explanation:** Adding a symbol slightly increases complexity, but the password is still based on a common word with predictable substitutions. The tool recognizes this pattern and still classifies it as very weak.



The screenshot shows the 'Password Strength Testing Tool' interface. At the top, the title 'Password Strength Testing Tool' is displayed in large blue letters. Below it, a subtitle reads 'Think you have a strong password? Find out below.' A text input field contains the password 'Password1!' in red text. Below the input field, the text 'Your password strength' is followed by a red progress bar that is only about 10% full. To the right of the progress bar, a red button labeled 'Very Weak' is visible. At the bottom, the text 'Estimated time to crack : 2 Seconds' is displayed.

### ❖ Tr0ub4dor&3

- **Strength Rating:** Good
- **Estimated Crack Time:** 4 months
- **Explanation:** This password includes uppercase and lowercase letters, numbers, and a symbol, significantly improving its strength. However, it is still derived from a recognizable word with substitutions, which reduces its overall security.

**Password Strength Testing Tool**

Think you have a strong password? Find out below.

Evaluate your password:

Tr0ub4dor&3

Your password strength **Good**

Estimated time to crack : **4 Months**

### ❖ k9!Qz\$L2@Wf8

- **Strength Rating:** Good
- **Estimated Crack Time:** 3 years
- **Explanation:** This password appears random and does not contain dictionary words or obvious patterns. The mix of character types and randomness increases entropy, making it far more resistant to brute-force attacks.

**Password Strength Testing Tool**

Think you have a strong password? Find out below.

Evaluate your password:

k9!Qz\$L2@Wf8

Your password strength **Good**

Estimated time to crack : **3 Years**

❖ **mR7\$A!2Z#9cL@4Q**

- **Strength Rating:** Strong
- **Estimated Crack Time:** Centuries
- **Explanation:** The password is long, fully random, and uses all character types. There are no detectable patterns, which dramatically increases the number of possible combinations attackers must try.

The screenshot shows the 'Password Strength Testing Tool' interface. At the top, the title 'Password Strength Testing Tool' is displayed in large white letters. Below it, a subtitle reads 'Think you have a strong password? Find out below.' A text input field contains the password 'mR7\$A!2Z#9cL@4Q'. Below the input field, the text 'Your password strength' is followed by a blue progress bar that is nearly full. To the right of the progress bar, a blue button labeled 'Strong' is visible. At the bottom, the text 'Estimated time to crack : Centuries' is displayed.

**Screenshot 7: correct-horse-battery-staple@37458**

- **Strength Rating:** Strong
- **Estimated Crack Time:** Centuries
- **Explanation:** This long passphrase combines multiple words with symbols and numbers. Its length provides very high entropy while remaining memorable, demonstrating why passphrases are considered one of the strongest password strategies.

The screenshot shows the 'Password Strength Testing Tool' interface. At the top, the title 'Password Strength Testing Tool' is displayed in large white letters. Below it, a subtitle reads 'Think you have a strong password? Find out below.' A text input field contains the passphrase 'correct-horse-battery-staple@37458'. Below the input field, the text 'Your password strength' is followed by a blue progress bar that is nearly full. To the right of the progress bar, a blue button labeled 'Strong' is visible. At the bottom, the text 'Estimated time to crack : Centuries' is displayed.

The following results are taken directly from the password strength testing tool screenshots provided. These reflect **actual tool feedback**, including strength labels and estimated crack times.

| # | Password Tested                    | Tool Strength Rating | Estimated Time to Crack | Explanation                                       |
|---|------------------------------------|----------------------|-------------------------|---------------------------------------------------|
| 1 | password                           | Very Weak            | Less than a second      | Dictionary word, no complexity, extremely common  |
| 2 | Password1                          | Very Weak            | Less than a second      | Predictable capitalization and number pattern     |
| 3 | Password1!                         | Very Weak            | 2 seconds               | Symbol added, but still follows a common pattern  |
| 4 | Tr0ub4dor&3                        | Good                 | 4 months                | Mixed characters and length, but still word-based |
| 5 | k9!Qz\$L2@Wf8                      | Good                 | 3 years                 | Randomized characters, no dictionary words        |
| 6 | mR7\$A!2Z#9cL@4Q                   | Strong               | Centuries               | Long, fully random, high entropy                  |
| 7 | correct-horse-battery-staple@37458 | Strong               | Centuries               | Very long passphrase with symbols and numbers     |

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## Best Practices for Creating Strong Passwords

1. **Length matters most** – Aim for **12–16+ characters**.
2. **Use all character types** – Uppercase, lowercase, numbers, symbols.
3. **Avoid dictionary words & names** – Even with substitutions.
4. **Avoid patterns** – No seasons, years, keyboard walks, or repeats.

5. **Prefer randomness or passphrases** – High entropy is key.
6. **Unique per account** – Never reuse passwords.
7. **Use a password manager** – Enables long, random passwords

## Tips Learned from the Evaluation

- Adding a symbol to a weak password helps **only slightly**.
  - Predictable substitutions (@ for a, 3 for e) are easily detected.
  - Long passphrases can outperform short complex passwords.
  - Randomness beats complexity tricks.
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## Common Password Attacks

### 1. Brute Force Attack

- Tries every possible combination.
- **Defense:** Long passwords and mixed character sets increase search space exponentially.

### 2. Dictionary Attack

- Uses common words and leaked password lists.
- **Defense:** Avoid real words; use randomness or passphrases.

### 3. Credential Stuffing

- Uses leaked passwords on multiple sites.
- **Defense:** Unique passwords + MFA.

### 4. Rainbow Table Attack

- Uses precomputed hashes.
  - **Defense:** Strong hashing + salts (system-side), long passwords (user-side).
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## Impact of Password Complexity on Security

- Higher complexity and length = higher entropy.
  - Increased entropy makes brute force attacks computationally expensive.
  - Complex, unique passwords reduce risk from automated attacks.
  - Passwords combined with multi-factor authentication (MFA) provide the strongest protection.
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## **Conclusion**

Password strength is determined by length, randomness, and character diversity. Testing passwords with strength checkers demonstrates that long, unique, and random passwords provide the best defense against common password attacks.